

Exp. 5(BCS551)-Subquery and Nested Query

Objective: At the end of this experiment, participants will be able to understand the concept and application of **Subqueries and Nested Queries** in SQL. They will learn how to retrieve data using **single-row and multiple-row subqueries**, apply **correlated subqueries**, and utilise **subqueries within SELECT, FROM, and WHERE clauses** to efficiently perform complex data retrieval and filtering operations.

Using a SubQuery:

Query1:

```
Select last_name, salary
FROM hr.employees
where salary >
(SELECT salary
from hr.EMPLOYEES
where last_name ='Abel');
```

Output1:

	LAST_NAME	SALARY
1	King	24000
2	Yang	17000
3	Garcia	17000
4	Gruenberg	12008
5	Singh	14000
6	Partners	13500
7	Errazuriz	12000
8	Ozer	11500

Executing Single-Row Subqueries:

Query2:

```
SELECT last_name, job_id, salary
FROM hr.employees
WHERE job_id = (
    SELECT job_id
    FROM hr.employees
    WHERE employee_id = 141
)
AND salary > (
    SELECT salary
    FROM hr.employees
    WHERE employee_id = 143
);
```

Output2:

	LAST_NAME	JOB_ID	SALARY
1	Nayer	ST_CLERK	3200
2	Mikkilineni	ST_CLERK	2700
3	Bissot	ST_CLERK	3300
4	Atkinson	ST_CLERK	2800
5	Mallin	ST_CLERK	3300
6	Rogers	ST_CLERK	2900
7	Ladwig	ST_CLERK	3600
8	Stiles	ST_CLERK	3200
9	Seo	ST_CLERK	2700
10	Rajs	ST_CLERK	3500

Query3:

```
SELECT last_name, job_id, salary
FROM hr.employees
WHERE salary = (
    SELECT MIN(salary)
    FROM hr.employees
);
```

Output3:

	LAST_NAME	JOB_ID	SALARY
1	Olson	ST_CLERK	2100

Query4:

```
SELECT department_id, MIN(salary)
FROM hr.employees
Group by department_id
Having MIN(salary) > (
    SELECT MIN(salary)
    FROM hr.employees
    Where department_id=50
);
```

Output4:

	DEPARTMENT_ID	MIN(SALARY)
1	90	17000
2	60	4200
3	100	6900
4	30	2500
5	80	6100
6	(null)	7000
7	10	4400
8	20	6000
9	40	6500
10	70	10000

Query5:

```
SELECT last_name, job_id
FROM hr.employees
WHERE job_id = (
    SELECT job_id
    FROM hr.employees
    WHERE last_name = 'Haas'
);
```

Output5:

LAST_NAME	JOB_ID
No items to display.	

Query6:

```
SELECT employee_id, last_name, job_id, salary
FROM hr.employees
WHERE salary < ANY (
    SELECT salary
    FROM hr.employees
    WHERE job_id = 'IT_PROG'
)
AND job_id <> 'IT_PROG';
```

Output6:

Download ▾ Execution time: 0.009 seconds					
	EMPLOYEE_ID	LAST_NAME	JOB_ID	SALARY	
1	132	Olson	ST_CLERK	2100	
2	136	Philtanker	ST_CLERK	2200	
3	128	Markle	ST_CLERK	2200	
4	135	Gee	ST_CLERK	2400	
5	127	Landry	ST_CLERK	2400	
6	191	Perkins	SH_CLERK	2500	
7	182	Sullivan	SH_CLERK	2500	
8	144	Vargas	ST_CLERK	2500	
9	140	Patel	ST_CLERK	2500	
10	131	Marlow	ST_CLERK	2500	

Using the ALL operator in Multiple-Row Subqueries

Query7:

```
SELECT employee_id, last_name, job_id, salary
FROM hr.employees
WHERE salary < ALL (
    SELECT salary
    FROM hr.employees
    WHERE job_id = 'IT_PROG'
)
AND job_id <> 'IT_PROG';
```

Output7:

  Download ▼ Execution time: 0.006 seconds				
	EMPLOYEE_ID	LAST_NAME	JOB_ID	SALARY
1	185	Bull	SH_CLERK	4100
2	192	Bell	SH_CLERK	4000
3	193	Everett	SH_CLERK	3900
4	188	Chung	SH_CLERK	3800
5	137	Ladwig	ST_CLERK	3600
6	189	Dilly	ST_CLERK	3600
7	141	Rajs	ST_CLERK	3500
8	186	Dellinger	SH_CLERK	3400
9	133	Mallin	ST_CLERK	3300
10	129	Bissot	ST_CLERK	3300

Query8:

```
Select emp.last_name
FROM hr.employees emp
where emp.employee_id NOT IN
(
  SELECT mgr.MANAGER_ID
  from hr.employees mgr
);
```

Output8:

LAST_NAME
No items to display.